

Scientific Computation

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Lecture 27 - Integration Error and Simpson's Rule

Lecture Outline

- Integration error

- Midpoint rule
- Simpson's rule

Integration Error

The trapezoidal rule amounts to approximating $f(x)$ by linear interpolation with

$$f(x) \approx p_1(x) = f(a) \frac{x-b}{a-b} + f(b) \frac{x-a}{b-a},$$

and then exactly computing the integral of the interpolant. We learned that the error of the degree n interpolant is

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i),$$

which for $n = 1$ gives us

$$f(x) - p_1(x) = \frac{f''(\xi(x))}{2!} (x-x_0)(x-x_1)$$

$$\text{for } \xi \in (a, b)$$

Integration Error for the Trapezoidal Method

The integration error is therefore

$$\int_a^b (f(x) - p_1(x)) dx = \overbrace{\int_a^b f(x) dx}^{I_f} - \overbrace{\frac{b-a}{2} [f(a) + f(b)]}^{I_{\text{trap}}}$$

$$= \frac{1}{2} \int_a^b f''(\xi(x)) (x-a)(x-b) dx$$

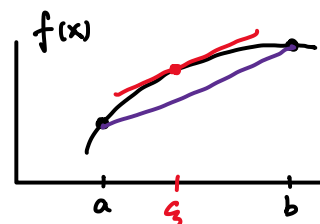
How do we evaluate this definite integral?

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The Mean-Value Theorem

Mean Value Theorem: For a function f that is continuous on $[a, b]$, there exists $\xi \in (a, b)$ such that

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$



Integral form: Let $F(x) \triangleq \int_a^x f(t) dt$

for continuous f on $[a, b] \Rightarrow F(x)$ is continuous, and we apply the MVT to $F(x)$: $\exists \xi \in (a, b)$, s.t. $F'(\xi) = f(\xi) = \frac{F(b) - F(a)}{b - a}$

$\exists \xi$ s.t.

$$f(\xi) (b-a)$$

$$= \frac{1}{b-a} \left(\int_a^b f(x) dx - \int_a^a f(x) dx \right)$$

$$= \int_a^b f(x) dx$$

$$= \frac{1}{b-a} \int_a^b f(x) dx$$

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Another MVT for Integrals

(Second) Mean Value Theorem for Integrals: Let f be continuous on $[a, b]$, and assume that g is a Riemann-integrable function on $[a, b]$. If $\underbrace{g(x) \geq 0}$, then there exists $\xi \in [a, b]$ such that

$$f \rightarrow -f, g \rightarrow 1$$

$$\hookrightarrow \text{or } g(x) \leq 0$$

$$\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx.$$

Think of it as a "positively weighted" integral: the simple form of the last slide is the special case where $g(x)=1$.

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Applying the Mean-Value Theorem

Note that $(x-a)(x-b) \leq 0$ does not change sign on $[a, b]$. Therefore, in order to apply the integral form of the MVT (with $g(x) = -(x-a)(x-b)$), we only need that $f''(\xi(x))$ is a continuous function of x on $[a, b]$ (see Exercise 4.2.12 in Cheney and Kincaid). Assuming this fact, we have

$$\frac{1}{2} \int_a^b f''(\xi(x)) (x-a)(x-b) dx = \frac{1}{2} f''(\xi) \int_a^b (x-a)(x-b) dx$$

where $y = \xi(v) \in (a, b)$, $v \in [a, b]$

using $\int_a^b (x-a)(x-b) dx = \frac{-(b-a)^3}{6}$, we have: $I_f - I_{\text{trap}} = \frac{-f''(y)}{12} \underbrace{(b-a)^3}_{\text{interval length}}$

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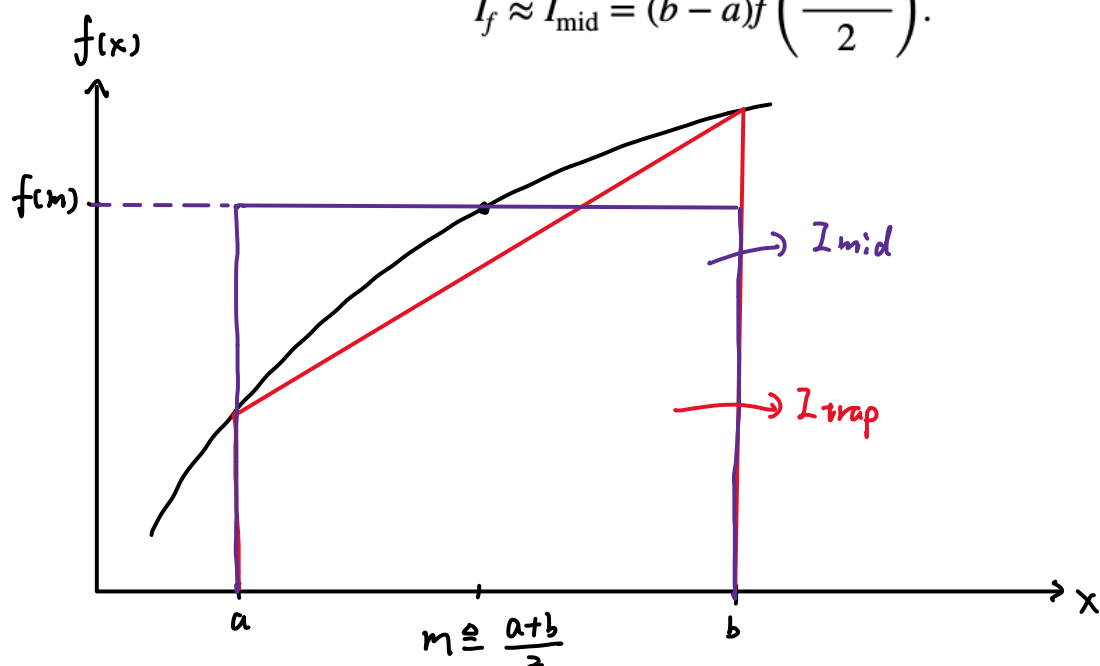
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The Midpoint Rule

Let's consider the case where we only evaluate $f(x)$ at the **midpoint** between $[a, b]$:

$$I_f \approx I_{\text{mid}} = (b - a)f\left(\frac{a + b}{2}\right).$$



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Error of the Midpoint Rule

Let $m = (a + b)/2$. Write the Taylor series

$$f(x) = f(m) + \underbrace{f'(m)(x - m)}_{0 \text{ in our integral}} + \underbrace{\frac{1}{2}f''(\xi(x))(x - m)^2}_{\bar{E}_2},$$

where $\xi(x)$ lies between x and m . Then we have

$$I_f = \int_a^b f(x) dx = (b - a)f(m) + \frac{1}{2} \int_a^b f''(\xi(x))(x - m)^2 dx$$

$$g(x) \triangleq (x - m)^2 \geq 0$$

$$\underbrace{I_f - I_{\text{mid}}}_{\text{Int. error}} = \frac{1}{2} \int_a^b f''(\xi(x))(x - m)^2 dx$$

Apply MVT Again for the Midpoint Rule

Since $(x - m)^2 \geq 0$, we can apply the same integral MVT to get

$$\frac{1}{2} \int_a^b f''(\xi(x)) (x-m)^2 dx = \frac{1}{2} f''(\eta) \int_a^b (x-m)^2 dx = \frac{f''(\eta)}{24} (b-a)^3$$

once again, we assume

$f''(\xi(x))$ is cont. with x

$$I_f - I_{mid} = \frac{f''(\eta) (b-a)^3}{24}$$

Comparing with the Trapezoidal Rule

Recall the error we derived for I_{trap} :

$\eta \neq \eta$ for trap and mid.

$$I_f - I_{trap} = \frac{-f''(\eta) (b-a)^3}{12} \xrightarrow{\text{red arrow}} = -2 (I_f - I_{mid})$$

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- **Simpson's rule**

**Higher Order Lagrange Basis
Polynomials**

Let's return to the setting where we approximate with a polynomial interpolant:

$$I_f = \int_a^b f(x)dx \approx \int_a^b p_n(x)dx.$$

We chose to write our interpolating polynomial as

$$p_n(x) = \sum_{j=0}^n f(x_j)L_j(x),$$

where $L_j(x)$ is the j -th Lagrange basis polynomial.

Setting $n=1$ give us the trapezoidal rule, we also (implicitly) assumed

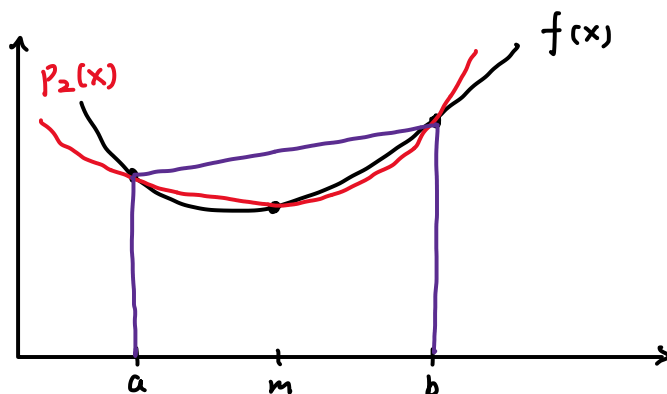
$$x_0 = a, \quad x_1 = b$$

Simpson's Rule

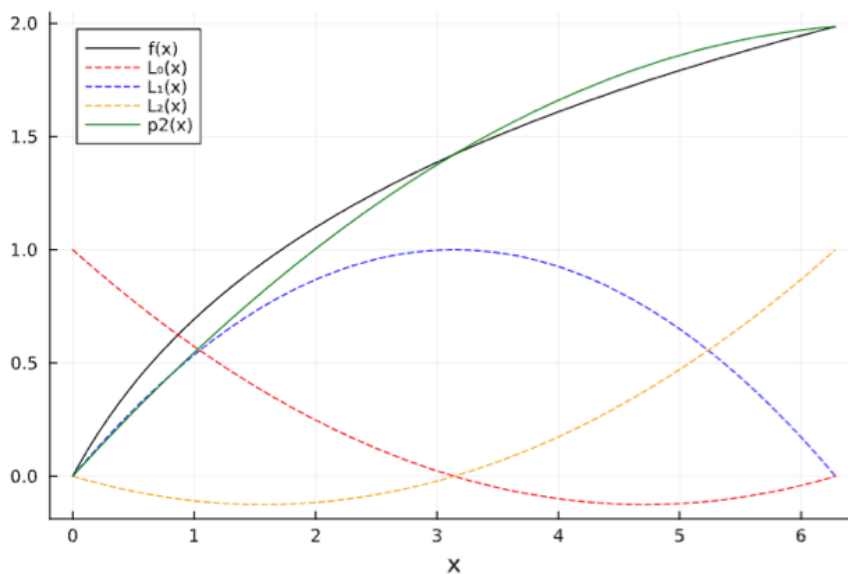
Let's try setting $n = 2$ and using $x_0 = a$, $x_1 = (a + b)/2$, and $x_2 = b$. We get

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x)$$

$$p_2(x) = f(a)L_0(x) + f(m)L_1(x) + f(b)L_2(x)$$



Simpson's Rule Visualized



```
using Plots
a = 0.0
b = 2*pi
m = (a+b)/2

f(x) = log(x+1.0)

x = a:0.01:b

L0(x) = (x - m)*(x - b)/((a-m)*(a - b))
L1(x) = (x - a)*(x - b)/((m-a)*(m - b))
L2(x) = (x - a)*(x - m)/((b-a)*(b - m))

p2(x) = f(a)*L0(x) + f(m)*L1(x) + f(b)*L2(x)

# err(η) = (b-a)^3/(η+1)/12

plot(x, f(x), label="f(x)", color=:black)
plot!(x, L0(x), label="L0(x)", color=:red,
       linestyle=:dash)
plot!(x, L1(x), label="L1(x)", color=:blue,
       linestyle=:dash)
plot!(x, L2(x), label="L2(x)", color=:orange,
       linestyle=:dash)
plot!(x, p2(x), label="p2(x)", color=:green)
# plot!(x, err(x), label="Error", color=:purple)
# hline!([f(a) + f(b)]/2, label="I_f/(b - a)")
xaxis!("x")
```

Integrating for Simpson's Rule

Let's evaluate the quadratic integrals which define Simpson's rule:

$$\int_a^b p_2(x) dx = f(a) \int_a^b L_0(x) dx + f\left(\frac{a+b}{2}\right) \int_a^b L_1(x) dx + f(b) \int_a^b L_2(x) dx$$

$$\int_a^b L_1(x) dx = \frac{1}{(m-a)(m-b)} \int_a^b (x-a)(x-b) dx = \frac{1}{(m-a)(m-b)} \int_a^b (x^2 - ax - bx + ab) dx$$

$\int_a^b \left(\frac{1}{3}x^3 - mx^3 + abx \right) \Big|_a^b$
 $\Rightarrow$ After lots of algebra, and repeating for L_0, L_2 :

$$I_{\text{Simpson}} = \frac{b-a}{6} [f(a) + 4f(m) + f(b)]$$

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Integrating the Exponent

Example 1: Compute $\int_0^1 e^x dx = e - 1 \approx 1.71828\dots$ once more, this time with Simpson's rule.

$$I_{\text{Simpson}} = \frac{1}{6} [f(0) + 4f(0.5) + f(1)]$$

$$= \frac{1}{6} [1 + 4e^{0.5} + e]$$

$$= 1.71886 \rightarrow \text{much more accurate from trapezoid rule!}$$

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Quadrature Summary

We have looked at three basic schemes with the following quadrature rules and expressions for error:

Quadrature	Integral	Error
Midpoint	$(b - a)f(m)$	$\frac{f''(\eta)}{24}(b - a)^3$
Trapezoidal	$\frac{(b - a)}{2}[f(a) + f(b)]$	$-\frac{f''(\eta)}{12}(b - a)^3$
Simpson	$\frac{(b - a)}{6}[f(a) + 4f(m) + f(b)]$	$-\frac{f^{(4)}(\eta)}{90}\left(\frac{b - a}{2}\right)^5$

这节课主要比较了三种数值积分方法和他们的误差；

1. 梯形法则的误差 (Trapezoidal Rule Error)

我们首先分析了梯形法则 的误差。

- **核心思想：** 梯形法则是用一个**一次多项式（直线）** $p_1(x)$ 来近似 $f(x)$ 。因此，“积分的误差” $\int (f(x) - p_1(x))dx$ 完全取决于“插值的误差” $f(x) - p_1(x)$ 。
- **推导：**
 1. 我们从 $n = 1$ 的**插值误差公式**开始： $f(x) - p_1(x) = \frac{f''(\xi(x))}{2}(x - a)(x - b)$ 。
 2. 总积分误差就是对这个式子进行积分： $Error_{trap} = \int_a^b \frac{f''(\xi(x))}{2}(x - a)(x - b)dx$ 。
- **难题：** 我们没法直接积分 $f''(\xi(x))$ ，因为它是一个我们不知道的、随 x 变化的函数。

2. 关键工具：(第二) 积分中值定理

为了解决上面的难题，我们引入了一个强大的工具：**(第二) 积分中值定理**。

- **定理：** $\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx$ 。
- **条件：** 这个定理能使用的**关键条件**是，“权重函数” $g(x)$ 在积分区间 $[a, b]$ 上**不能改变符号**（即始终 > 0 或始终 < 0 ）。

- **应用：**这正是我们需要的！它允许我们把那个“棘手的” $f(x)$ （在我们的例子中是 f'' ）从积分号里“拉”出来，变成一个常数 $f(\xi)$ 。

3. 误差公式的最终推导

我们使用这个 MVT 工具分别解出了两个误差公式：

梯形法则 (Trapezoidal)

1. **检查：**权重 $g(x) = (x - a)(x - b)$ 在 $[a, b]$ 上始终 ≤ 0 。条件满足。
2. **应用：** $Error_{trap} = \frac{1}{2}f''(\xi_y) \int_a^b (x - a)(x - b)dx$ 。
3. **求解：**简单的积分 $\int_a^b (x - a)(x - b)dx = -\frac{(b-a)^3}{6}$ 。
4. **结果：** $Error_{trap} = \frac{-f''(\xi_y)}{12}(b - a)^3$ 。

中点法则 (Midpoint)

1. **推导：**我们使用泰勒展开来推导中点法则的误差，得到了一个不同的误差积分： $Error_{mid} = \frac{1}{2} \int_a^b f''(\xi(x))(x - m)^2 dx$ 。
2. **检查：**权重 $g(x) = (x - m)^2$ 在 $[a, b]$ 上始终 ≥ 0 。条件满足。
3. **应用：** $Error_{mid} = \frac{1}{2}f''(\xi_y) \int_a^b (x - m)^2 dx$ 。
4. **求解：**简单的积分 $\int_a^b (x - m)^2 dx = \frac{(b-a)^3}{12}$ 。
5. **结果：** $Error_{mid} = \frac{f''(\xi_y)}{24}(b - a)^3$ 。

4. 比较与推广

- **比较：**
 - $Error_{trap} \approx \frac{-1}{12}(\dots)$
 - $Error_{mid} \approx \frac{+1}{24}(\dots)$
 - **结论：**中点法则的精度通常是梯形法则的两倍，并且它们的误差符号相反。
- **推广（辛普森法则）：**
 - 我们最后将积分与拉格朗日插值统一起来。
 - 梯形法则被揭示为** $n = 1$ （线性）**的拉格朗日插值积分。
 - 这启发我们使用 $n = 2$ （抛物线）的拉格朗日插值，它使用 a, b 和中点 m 这 3 个点。
 - 通过精确地积分这个 $n = 2$ 的插值多项式 $P_2(x)$ ，我们推导出了一个更高级、更精确的公式：**辛普森法则 (Simpson's Rule)**。